Weighted Modal Logic in Epistemic and Deontic Contexts

Huimin Dong[1], Xu Li[2], and Yì N. Wáng[1(✉)]

[1] Department of Philosophy (Zhuhai), Sun Yat-sen University, Zhuhai, China
{huimin.dong,ynw}@xixilogic.org
[2] Department of Computer Science, University of Luxembourg, Esch-sur-Alzette, Luxembourg
xu.li@uni.lu

Abstract. We introduce a type of weighted modal logic with explicit weights both in the language and in the models. The framework has its applications in epistemic logic for reasoning about agents' knowledge based on their capability, and in deontic logic for agents' choices based on their deontic capability or utilities. We make use of weighted Kripke models with the weights understood epistemically as a similarity measure between states and deontically as a measure of expected utilities. We present sound and complete axiomatizations for the logics, and discuss variants and possible extensions.

Keywords: Weighted modal logic · Epistemic logic · Deontic logic · Epistemic capability · Deontic capability

1 Introduction

In standard epistemic logic [5,14,20], whether an agent knows a formula is interpreted as whether the formula is true in all states the agent considers possible. A binary relation over the set of states is then used for modeling the indistinguishability relation of an agent. In this way of modeling, in order to verify the truth of an epistemic formula $K_a\varphi$ (reads "a knows φ") or $\hat{K}_a\varphi$ (reads "a considers φ possible"), it is important to check whether φ is true in *all* or *some* of the states that a cannot distinguish from the factual state. Neither the number of these indistinguishable states matters, nor is there any difference in the level of relevance or importance of these states.

When we start to enhance the logic to make it able to reason about an agent's capability of discerning between states, we find it useful to somehow model the *quantity* or *quality* of states. There are frameworks studied in the literature that are relevant to this topic. For example, in graded modal logic [6,7], in particular its applications to epistemic logic [15], graded modal operators can be used to characterize different *degrees* (or *levels of unconfidence*) of knowledge, where "n-degree knowledge" (when n goes higher the agent is more unconfident) of φ is interpreted formally as having no more than n $\neg\varphi$-states that are indistinguishable from the factual state. Another way of incorporating the quantity is

© Springer Nature Switzerland AG 2021
S. Ghosh and T. Icard (Eds.): LORI 2021, LNCS 13039, pp. 73–87, 2021.
https://doi.org/10.1007/978-3-030-88708-7_6

adopted in weighted modal logic (or weighted transition systems) [9,18], where weights are added to the binary relation to model the costs of transitions. Put in the context of epistemic logic, the weight can be understood to be the difference or distance between two states [22]. "n-degree knowledge" of φ, in this case, can be interpreted as φ being true in all states that are different from the factual state in the distance of up to n – this time the bigger n goes the more confident the agent is.

In this paper, we do not limit our eyesight in the scope of an epistemic context. We study a general type of multi-agent weighted modal logics in which the weights are partially ordered. Its formal semantics is based on weighted Kripke models. We introduce a sound and complete proof system for the minimal logic. The framework is flexible enough for modeling scenarios from different perspectives, including an epistemic one.

In particular, we model agents' capability by interpreting the degree of knowledge using weighted models, however, in terms of *similarity* between states, other than their distance or the quantity of states. A formula $K_a^r\varphi$ will read "a knows φ with her level of capability set to r", and be formally interpreted as "φ is true in all states that are similar, to the degree at least r, to the factual state". The idea of measuring similarity between states is not new, which can be traced back to at least [19, Section 2.4].

Another application of the logic is in a deontic context. We can treat the weight to be the quantity or quality of an agent's deontic capability or utility. A formula $O_a^r\varphi$ reads "φ is obligatory for a under the moral standard r". Semantically, $O_a^r\varphi$ is true in a state if φ is true all alternative states that give the degree of a's utilities no less than r.

The structure of the paper is as follows. In Sect. 2 we introduce the minimal logic and its sound and complete axiomatization. Its applications in epistemic and deontic contexts are studied in Sects. 3 and 4, respectively. We discuss further topics and conclude in Sect. 5.

2 Minimal Logic

In this section we introduce a weighted multi-agent modal logic with very weak constraints on the weights. This will serve as a general framework for modeling scenarios from an epistemic or deontic perspective.

2.1 Syntax and Semantics

Let Ag denote the set of all agents and $Prop$ the set of all propositional variables. As usual, we assume Ag to be a finite set, and $Prop$ a countably infinite set. Furthermore, we use a partially ordered (nonempty) set, $(I, \preceq)$, to characterize the weights (relativity or degree) of modal operators. Given $r_1, r_2 \in I$, we sometimes write $r_1 \succeq r_2$ for $r_2 \preceq r_1$ for convenience. The syntax and semantics are dependent on these parameters.

Definition 1 (language). *The language $\mathcal{L}$ is given by the following grammar rule:*

$$\varphi ::= p \mid \neg\varphi \mid (\varphi \to \varphi) \mid \Box_a^r \varphi$$

where $p \in Prop$, $a \in Ag$ and $r \in I$. We define $\Diamond_a^r \varphi$ as $\neg\Box_a^r\neg\varphi$. Other Boolean connectives are defined as usual.

Definition 2 (models). *A* model *for $\mathcal{L}$ is a tuple $M = (W, R, V)$ such that*

- *W is a non-empty set, whose elements are called worlds or states;*
- *$R : Ag \to (W \times W) \to I$ is a labeling function that assigns a weight to every pair of worlds for every agent;*
- *$V : Prop \to \wp(W)$ is a valuation.*

Definition 3 (satisfaction). *Given a model $M = (W, R, V)$, a world $w \in W$, and a formula φ, the satisfaction of φ in (M, w) (notation: $M, w \models \varphi$) is defined inductively as follows:*

$$\begin{array}{lll} M, w \models p & \iff & w \in V(p) \\ M, w \models \neg\psi & \iff & \textit{not } M, w \models \psi \\ M, w \models (\psi \to \chi) & \iff & M, w \models \psi \textit{ materially implies } M, w \models \chi \\ M, w \models \Box_a^r \psi & \iff & M, u \models \psi \textit{ for all } u \textit{ such that } R_a(w, u) \succeq r. \end{array}$$

Validity and logical consequence can be defined in a usual way (cf., say, [24]).

Proposition 1. *The following hold for all $a, b \in Ag$, $r, r_1, r_2 \in I$ and $\varphi, \psi \in \mathcal{L}$:*

1. *$\models \Box_a^{r_1}\varphi \to \Box_a^{r_2}\varphi$, if $r_1 \preceq r_2$;*
2. *$\models \Box_a^r(\varphi \to \psi) \to (\Box_a^r\varphi \to \Box_a^r\psi)$;*
3. *If $\models \varphi$, then $\models \Box_a^r\varphi$.*

We write $\min(I)$ for the set of minimal elements of I with respect to the partial order $\preceq$, i.e., $\min(I) = \{i \in I \mid \forall s \in I : s \preceq i \Rightarrow s = i\}$. We say that $(I, \preceq)$ is *based* on J (with $J \subseteq I$), if for all $i \in I$ there is $j \in J$ such that $j \preceq i$. We call $(I, \preceq)$ *has a base* if it is based on some $J \subseteq I$, and *finitely based* if it has a base that is finite.

Lemma 1. *If $(I, \preceq)$ had a base, then for any $i \in I$, there exists $i^* \in \min(I)$ such that $i^* \preceq i$.*

Proof. We show the lemma by proving the following stronger claim inductively:

For any partially ordered (nonempty) set $(J, \preceq)$ that has a base, for any $j \in J$, there exists $j^* \in \min(J)$ such that $j^* \preceq j$. (†)

Since $(J, \preceq)$ has a base, there exists a subset J' of J such that for all $j \in J$, there exists $j' \in J'$ with $j' \preceq j$. By the transitivity of $\preceq$ it suffices to show that for all $j' \in J'$ there exists $j^* \in \min(J)$ such that $j^* \preceq j'$. By the induction hypothesis, since $(J', \preceq)$ has a base, for any $j' \in J'$, there exists $j^{**} \in \min(J')$ such that $j^{**} \preceq j'$. By definition, $\min(J) = \min(J')$, and j^{**} can serve as the desired j^*. □

Proposition 2. *Let $(I, \preceq)$ be finitely based and let $[\mathbf{0}]_a\varphi$ denote $\bigwedge_{i\in\min(I)} \Box^i_a\varphi$. The following hold for every $a, b \in Ag$, $r, r_1, r_2 \in I$ and $\varphi, \psi \in \mathcal{L}$:*

1. $\models [\mathbf{0}]_a\varphi \to \varphi$;
2. $\models \neg[\mathbf{0}]_a\varphi \to [\mathbf{0}]_a\neg[\mathbf{0}]_a\varphi$;
3. $\models [\mathbf{0}]_a\varphi \leftrightarrow [\mathbf{0}]_b\varphi$.

2.2 Axiomatization

We first show that, when $(I, \preceq)$ is not finitely based, the minimal logic is not *compact*, i.e., there is a set Γ of $\mathcal{L}$ formulas which is unsatisfiable while all of its finite subsets are satisfiable.

Lemma 2. *If $(I, \preceq)$ is not finitely based, then the minimal logic is not compact.*

Proof. Suppose $(I, \preceq)$ is not finitely based. Consider a set of formulas $\Gamma = \{\Diamond^r_a\Diamond^r_a p\} \cup \{\Box^i_a\neg p \mid i \in I\}$. Γ must be infinite, for otherwise I is finite and $(I, \preceq)$ must be finitely based. We show that every finite subset of Γ is satisfiable but it is not the case with itself.

To see that Γ is unsatisfiable, suppose a model $M = (W, R, V)$ and $w \in W$ such that $M, w \models \varphi$ for each $\varphi \in \Gamma$. Since $M, w \models \Diamond^r_a\Diamond^r_a p$, there must be worlds $u, v \in W$ such that $R_a(w, u) \succeq r$, $R_a(u, v) \succeq r$ and $M, v \models p$. Now assume that $R_a(w, v) = i \in I$, it follows from the semantics that $M, w \models \Diamond^i_a p$, which contradicts the fact that $M, w \models \Box^i_a\neg p$ for all $i \in I$.

Now we show that every finite subset of Γ is satisfiable. Let $J \subseteq I$ be an arbitrary finite subset of I. By definition, if $(I, \preceq)$ is not finitely based, then there is $i \in I\backslash J$ such that $i \not\succeq x$ for all $x \in J$. Now we construct a model $M^\Delta = (W^\Delta, S^\Delta, V^\Delta)$ in the aim of showing that $M^\Delta, w \models \bigwedge \Delta$ for an arbitrary finite subset Δ of Γ (let $J_\Delta = \{i \in I \mid \Box^i_a\neg p \in \Delta\}$):

- $W^\Delta = \{w, u, v\}$;
- $S^\Delta_a(w, u) = R_a(u, v) = r$, $R_a(w, v) = i$ such that $i \in I\backslash J_\Delta$ and $i \not\succeq x$ for all $x \in J_\Delta$ (such i exists by $(*)$);
- $V^\Delta(p) = \{v\}$.

It is not hard to see that $M^\Delta, w \models \bigwedge \Delta$. □

For noncompact logics we will not have a strong complete axiomatization (cf., say, [24]), and in this paper we focus only on the cases when $(I, \preceq)$ *is finitely based*. We introduce a Hilbert-style axiom system **K** for the minimal logic, which is presented in Fig. 1.

The validity of the axioms (UM) which reads "upward monotonicity", (K), (T^0), (5^0) and (A^0), and the rule (N) is found in Propositions 1 and 2. We can easily achieve the following result.

Theorem 1 (soundness). **K** *is sound for the minimal logic, under the condition that $(I, \preceq)$ is finitely based.*

(**PC**)	all instances of propositional tautologies
(UM)	$\Box_a^{r_1}\varphi \to \Box_a^{r_2}\varphi$, if $r_1 \preceq r_2$
(K)	$\Box_a^r(\varphi \to \psi) \to (\Box_a^r\varphi \to \Box_a^r\psi)$
(T^0)	$[\mathbf{0}]_a\varphi \to \varphi$
(5^0)	$\neg[\mathbf{0}]_a\varphi \to [\mathbf{0}]_a\neg[\mathbf{0}]_a\varphi$
(A^0)	$[\mathbf{0}]_a\varphi \leftrightarrow [\mathbf{0}]_b\varphi$
(MP)	from φ and $\varphi \to \psi$, infer ψ
(N)	from φ infer $\Box_a^r\varphi$

Fig. 1. The axiom system **K**, in which $a, b \in Ag$, r, r_1 and r_2 are from a partially ordered set $(I, \preceq)$, and φ and ψ are formulas of $\mathcal{L}$.

We now move on to the completeness of **K**. The proof is essentially by the unraveling method [25], but in terms of [29] for a shorter proof.

We denote the set of all maximal consistent sets of formulas by MCS.

Definition 4 (canonical relation). *The* canonical relation $\mathcal{R}_a^r$ *for a given agent a and a given $r \in I$ is a binary relation over MCS, such that for any $\Phi, \Psi \in MCS$, $(\Phi, \Psi) \in \mathcal{R}_a^r$ iff for all formulas φ, $\Box_a^r\varphi \in \Phi$ implies $\varphi \in \Psi$.*

Definition 5 (canonical paths). *Given $\Psi \in MCS$, a* canonical path *w starting from Ψ is a finite sequence $\langle \Phi_0, \mathcal{R}_{a_1}^{r_1}, \Phi_1, \mathcal{R}_{a_2}^{r_2}, \Phi_2, \ldots, \mathcal{R}_{a_n}^{r_n}, \Phi_n \rangle$ such that all the following hold:*

- *$\Psi = \Phi_0$;*
- *For all $i = 0, \ldots, n$, $\Phi_i \in MCS$;*
- *For all $i = 1, \ldots, n$, $a_i \in Ag$ and $r_i \in I$;*
- *For all $i = 0, \ldots, n-1$, $\mathcal{R}_{a_i}^{r_i}$ is a canonical relation for a_i and r_i s.t. $(\Phi_i, \Phi_{i+1}) \in \mathcal{R}_{a_{i+1}}^{r_{i+1}}$.*

We simply call w a canonical path *if the starting element is not important in the context. We write $tail(w)$ for the last element of the canonical path, which is Φ_n in the above case.*

Lemma 3. *Let w and u be two canonical paths starting from Ψ. Then for any agent $a \in Ag$, there exists $r \in \min(I)$ such that $(tail(w), tail(u)) \in \mathcal{R}_a^r$.*

For any canonical paths w, u starting from Ψ and any $a \in Ag$, let $D_a(w, u) = \{r \in \min(I) \mid (tail(w), tail(u)) \in \mathcal{R}_a^r\}$. By the above lemma, $D_a(w, u)$ is not empty. We choose an arbitrary $d_a(w, u)$ from $D_a(w, u)$ and fix it for the use below.

Definition 6 (standard model). *The* standard model *for a given $\Psi \in MCS$ is a tuple $\mathsf{M} = (W, R, V)$ such that:*

1. *W is the set of all canonical paths starting from Ψ,*
2. *For any $w, u \in W$ and any $a \in Ag$,*

$$R_a(w,u) = \begin{cases} r, & \text{if } u \text{ extends } w \text{ with } \langle \mathcal{R}_a^r, \Phi \rangle \text{ for some } \Phi \in MCS, \\ d_a(w,u), & \text{otherwise.} \end{cases}$$

3. *For any propositional variable p, $V(p) = \{w \in W \mid p \in tail(w)\}$.*

Proposition 3 (standardness). *The standard model M for a given $\Psi \in MCS$ is a model.*

Lemma 4 (existence). *Given a standard model $\mathsf{M} = (W, R, V)$ for a given $\Psi \in MCS$, for any $r \in I$, any $a \in Ag$ and any $w \in W$ and any formula φ, if $\Box_a^r \varphi \notin tail(w)$, then there exists $u \in W$ such that $R_a(w,u) \succeq r$ and $\varphi \notin tail(u)$.*

Proof. Let $\Box_a^r \varphi \notin tail(w)$ as given. Let $\Phi = \{\neg\varphi\} \cup \{\psi \mid \Box_a^r \psi \in tail(w)\}$. We can show that Φ is consistent. Let Φ' be a maximal consistent set of formulas extending Φ. We have $(tail(w), \Phi') \in \mathcal{R}_a^r$. Let $u = \langle w, \mathcal{R}_a^r, \Phi' \rangle$. Since $R_a(w,u) = r$ and clearly $\varphi \notin tail(u)$, so u is a witness for the lemma. □

Lemma 5 (truth). *Let $\mathsf{M} = (W, R, V)$ be the standard model for a given $\Psi \in MCS$ and $w \in W$. For any formula φ, $\mathsf{M}, w \models \varphi$ if and only if $\varphi \in tail(w)$.*

Proof. Let $\mathsf{M} = (W, R, V)$ and $w \in W$ be given. The proof is by induction on φ. If φ is an atomic proposition, then it follows from Definitions 3 and 6. Boolean cases follow from maximality and consistency of the set $tail(w)$. It remains to deal with the modalities.

Suppose φ is of the form $\Box_a^r \psi$. The direction from left to right follows from the existence lemma and the induction hypothesis. From right to left, suppose $\Box_a^r \varphi \in tail(w)$. Let $u \in W$ be an arbitrary world such that $R_a(w,u) \succeq r$, it suffices to show that $M, u \models \varphi$. Since $\Box_a^r \varphi \in tail(w)$, it follows from the axiom (UM) that $\Box_a^{R_a(w,u)} \varphi \in tail(w)$. From the construction of R_a we can show that $(tail(w), tail(u)) \in \mathcal{R}_a^{R_a(w,u)}$. This involves some details which are omitted here. So $v \in tail(u)$. Applying the induction hypothesis, we obtain $M, u \models \varphi$ as we desired. □

Theorem 2 (Kcompleteness). *For any set of formulas Φ and any formula φ, if $\Phi \models \varphi$, then $\Phi \vdash_{\mathbf{K}} \varphi$.*

Proof. Suppose $\Phi \nvdash \varphi$, it follows that $\Phi \cup \{\neg\varphi\}$ is consistent. $\Phi \cup \{\neg\varphi\}$ can be extended to a maximal consistent set Ψ of formulas. Let M be the standard model for Ψ. Since $\langle \Psi \rangle$ is a canonical path, it is a world in M. By the truth lemma, for any formula ψ, $\mathsf{M}, \langle \Psi \rangle \models \psi$ iff $\psi \in \Psi$, by which it means Ψ is satisfiable. It follows that $\Phi \cup \{\neg\varphi\}$ is satisfiable, and so $\Phi \not\models \varphi$. □

3 Weights as Epistemic Capability

In this section, we interpret the weighted modal logic in an epistemic context. In this case, we specify the partially ordered set $(I, \preceq)$ as the closed interval $[0, 1]$

of the real line, with the order $\preceq$ understood as the ordinary relation $\leq$ (i.e., "less than or equal to").

Models for this kind of epistemic logic is a subclass of the models for the minimal logics (Definition 2). More precisely,

Definition 7 (epistemic models). *A tuple $M = (W, R, V)$ is called an* epistemic model *if it is such that:*

- *W is a non-empty set of epistemic states (or worlds);*
- *$R : Ag \to (W \times W) \to [0, 1]$ is a labeling function that assigns a weight (in the interval $[0, 1]$) for every pair of worlds for every agent a, such that the following properties hold for all $w, u, v \in W$:*
  (a) $R_a(w, u) = 1$ implies $w = u$;
  (b) $R_a(w, u) = R_a(u, w)$;
- *$V : Prop \to \wp(W)$ is a valuation.*

The labeling function R introduced in the definition above is intended to model the similarity between worlds for a given agent. In a multi-agent setting, similarity measurement can be subjective. Intuitively, for an agent a, given two worlds w and u, $R_a(w, u)$ tells the degree of similarity between w and u by a. Agents can hold different opinions toward worlds regarding how similar they are.

There are two constraints enforced on the similarity relation of an agent. We consider that (i) any two different worlds are not ultimately similar, and (ii) it is to the same degree that world w is similar to world u and that the other way round. These considerations are expressed by the constraints (a) and (b), respectively.

One may find the constraints we have chosen the same as those considered in the area of data mining (see, e.g., [26, Section 2.4.4]). For what we have done in this paper, it is fair for one to regard each world as a data object consisting of a number of attributes, and the similarity as the degree to which two worlds are alike. Similarity falls in the interval $[0, 1]$ with 0 representing no similarity and 1 complete similarity (all attributes of two data objects match).

One may in addition consider a third requirement (iii) a world is indistinguishable from (i.e., ultimately similar to) itself, which can be achieved by putting an extra constraint on R_a:

(c) $R_a(w, u) = 1$ if $w = u$;

which, together with (a), makes $R_a(w, u)$ iff $w = u$.

The formula $\Box_a^r$ is often written as $K_a^r\varphi$ in an epistemic context, expressing that "agent a knows with *capability* or *epistemic effort* r that φ is the case". The satisfaction relation is defined exactly the same as for the minimal logic (Definition 3). In this paper we call such a logic an *epistemic logic*, which differs from standard epistemic logic [5,20] in the power of expressing epistemic capability.

When $r = 1$, the formula $K_a^1\varphi$ expresses that φ is some kind of God's knowledge, i.e., the knowledge of an omniscient agent or someone with supreme power. In this case, every fact is known by the supreme agent, which is expressed by the

valid formula $\varphi \to K_a^1\varphi$. The supreme knowledge may not be consistent, though. When the current world is not ultimately similar to itself, say $R_a(w, w) = 0.8$, the supreme agent "knows" that w cannot be a factual world (and in fact there is no factual world in this case), and $K_a^1\psi$ is true in w for every formula ψ. This can be ruled out by adding the constraint (c) in the definition of an epistemic model, and this will enforce that the true formulas are exactly knowledge of the supreme agent (i.e., $\varphi \leftrightarrow K_a^1\varphi$ is valid in this way).

When $r = 0$, the formula $K_a^0\varphi$ says that φ is known by a grossly ignorant agent a, an idle agent without any effort employed. The agency means little in this kind of formulas, and technically K_a^0 falls back to the *universal modality*.

As an example for the model checking in the introduced logic, consider the model $M = (\{w, u, v\}, R, V)$ such that:

- $R_a(w, w) = R_a(v, v) = 1$
- $R_a(u, u) = 0.8$
- $R_a(w, u)$=$R_a(u, w) = 0.6$
- $R_a(w, v)$=$R_a(v, w) = 0.2$
- $R_a(u, v) = R_a(v, u) = 0.4$
- $V(p) = \{w\}$.

An illustration of M is given in Fig. 2. The truth values of some formulas in the model M are given below:

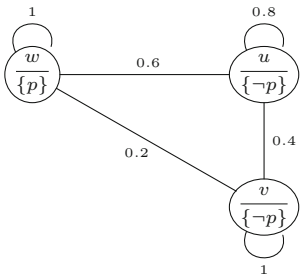


Fig. 2. Illustration of the model M.

- The formula $K_a^1 p$ is true at w and u, but false at v in the model M;
- The formula $K_a^{0.7}\neg p$ is false at w, and true at u and v in the model M;
- The formula $K_a^{0.5} p$ is false at w, u and v in the model M;
- The formula $K_a^{0.3}\neg p$ is false at w and u, and true at v in the model M;
- The formula $\neg K_a^0 p$ is true at w, u and v in the model M.

3.1 Axiomatization

All **K** axioms are valid in the class of all epistemic models, since the latter is a subset of the class of all models. It is also not hard to observe that all **K** rules still preserve validity. So **K** can be a subsystem for the logic over the class of epistemic models.

In the epistemic context, $\min(I) = \min([0,1]) = 0$, so the formula $[\mathbf{0}]_a\varphi$ is $K_a^0\varphi$. The characterization axioms for grossly ignorant agents, denoted (**Ign**), include axioms (T^0), (5^0) and (A^0). The characterization axioms for omniscient agents, denoted (**Omn**), include (Tr^1) and (T^1). These are listed in Figs. 3 and 4.

(T^0)	$K_a^0\varphi \to \varphi$
(5^0)	$\neg K_a^0\varphi \to K_a^0\neg K_a^0\varphi$
(A^0)	$K_a^0\varphi \leftrightarrow K_b^0\varphi$

Fig. 3. Characterization axioms (**Ign**).

(Tr^1)	$\varphi \to K_a^1\varphi$
(T^1)	$K_a^1\varphi \to \varphi$
	(not needed unless constraint (c) is enforced)

Fig. 4. Characterization axioms (**Omn**).

The axiom (UM) expresses the assumption that if a fact is known by some agent with capability to some degree, then it is also known when the agent's capability is greater.

We introduce an axiomatization **E** for the epistemic logic which is shown in Fig. 5. It is clear that **E** extends **K** with the axioms (B), (Tr^1) and (T^1).

It is not hard to verify the soundness of the axiomatization. For example, (Tr^1), (B) and (T^1) characterize the conditions (a), (b) and (c) of an epistemic model, respectively. We focus on the completeness of **E** here. The completeness proof can be carried out in the same way as that for **K**. Below we only note the differences.

We define *canonical relation* and *canonical paths* in the same way as we did for the minimal logic (Definitions 4 and 5).

Proposition 4. *In the epistemic logic, the canonical relation $\mathcal{R}_a^r$ is symmetric for any $a \in Ag$ and $r \in [0,1]$.*

(**PC**)	all instances of propositional tautologies
(UM)	$K_a^{r_1}\varphi \to K_a^{r_2}\varphi$, if $r_1 \leq r_2$
(K)	$K_a^r(\varphi \to \psi) \to (K_a^r\varphi \to K_a^r\psi)$
(B)	$\neg\varphi \to K_a^r\neg K_a^r\varphi$
(**Ign**)	characterization axioms for grossly ignorant agents
(**Omn**)	characterization axioms for omniscient agents
(MP)	from φ and $\varphi \to \psi$, infer ψ
(N)	from φ infer $K_a^r\varphi$

Fig. 5. The axiom system **E**.

In the definition of a *standard model* (see Definition 6 in the case of the minimal logic), we need to replace the constraint on R with the following:

$$R_a(w,u) = \begin{cases} 1, & \text{if } w = u \text{ and } \langle w, R_a^1, tail(w)\rangle \text{ is a canonical path,} \\ s, & \text{if } s \neq 1 \text{ and } w \text{ extends } u \text{ with } \langle \mathcal{R}_a^s \Phi\rangle \text{ for some } \Phi \in MCS, \\ s, & \text{if } s \neq 1 \text{ and } u \text{ extends } w \text{ with } \langle \mathcal{R}_a^s, \Phi\rangle \text{ for some } \Phi \in MCS, \\ 0, & \text{otherwise.} \end{cases}$$

We can then show the Lemmas of Standardness, Existence and Truth in a similar way, which leads to the following completeness result.

Theorem 3 (Ecompleteness). *For any set of formulas Φ and any formula φ, if $\Phi \models \varphi$, then $\Phi \vdash_{\mathbf{E}} \varphi$.*

4 Weights as Deontic Capability

In this section we use the weighted modality $\Box_a^r$ to capture agent a's *deontic capability* of fulfilling her desire. This modality of norms goes along with obligation in standard deontic logic [13], while explicitly stating the degree of deontic capability. Deontic capability can be taken as moral standard: an agent is in a higher degree of deontic capability of making a choice, when the agent has a higher moral standard to evaluate this choice. We write $O_a^r\varphi$ for $\Box_a^r\varphi$ in a deontic context, to express "agent a considers φ obligatory under the moral standard r". We leave the comparison with other approaches in deontic literature in Sect. 5.

We interpret this modality of deontic capability in terms of utility recognition. Semantically, a sentence $O_a^r\varphi$ manifests an agent's *ability of recognizing* the differences of various utilities. As in decision theory, the outcome of a choice is represented by utility. An agent upholds a moral standard r in achieving φ, if and only if φ is true in all possible choices (states) that the agent recognizes with utility no less than r.

For instance, a responsible seller considers the choices of protecting the customer's privacy resulting in a high utility. The choices recognized by a reckless agent in a lower utility include privacy exposure, and it is not obligatory for this reckless agent to protect personal information. In short, an agent has an r-degree

of deontic capability of making a choice if and only if this agent recognizes this choice brought in a utility no less than r. Now we make this formally.

Definition 8 (deontic models). *Given $\rho \in I$, a ρ-rank* deontic model *is a tuple $M = (W, R, V)$ such that:*

- *W is a non-empty set of worlds or states;*
- *$R : Ag \to (W \times W) \to I$ is a function such that:*

$$\textit{for all } r \preceq \rho, \textit{ there exists } u \in W \textit{ such that } R_a(w, u) \succeq r \qquad (\mathcal{C}^\rho)$$

- *$V : Prop \to \wp(W)$ is a valuation.*

In a deontic model (W, R, V) with $w, u \in W$, we can understand $R_a(w, u)$ as "agent a's utility of shifting to the state u from w". The truth condition of $O_a^r\varphi$ is defined just as in Sect. 2. Namely,

$$M, w \models O_a^r v \iff \text{for all } u \in W, \text{ if } R_a(w, u) \succeq r \text{ then } M, u \models \varphi.$$

The condition $(\mathcal{C}^\rho)$ indicates that, under any utility not higher than the rank ρ, there is an alternative for agent a to achieve φ. The condition $(\mathcal{C}^\rho)$ guarantees the validity of the formula $\neg(O_a^r\varphi \wedge O_a^r\neg\varphi)$ (for any $r \leq \rho$) which describes deontic consistency for agent a with her moral standard not higher than the rank ρ. Moreover, just as in the minimal logic, the formula $(O_a^{r_1}\varphi \to O_a^{r_2}\varphi)$ is valid whenever $r_1 \preceq r_2$. This validity states that, when a choice of φ is obligatory given that agent a's moral standard is r_1, this choice is still obligatory in case of a higher moral standard than r_1.

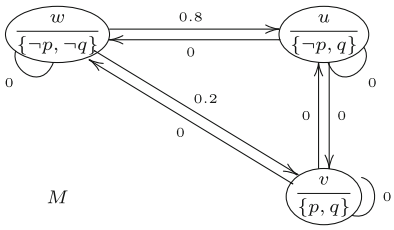
Note that $\preceq$ is a partial order which may not be linear, hence there can be incomparable utilities, i.e., there exist $r_1, r_2 \in I$ such that neither $r_1 \preceq r_2$ nor $r_2 \preceq r_1$. For simplicity, however, in what follows we focus on a linear order. In particular, here we regard the partially ordered set $(I, \preceq)$ as the closed interval $[0, 1]$ of real line, with the order $\preceq$ the standard relation $\leq$ which represents "less than or equal to". For example, when agent a is obligatory to achieve φ by demanding the utility not lower than 1, $O_a^1\varphi$ expresses that a has the highest capability or moral standard to produce the most good as φ. When one is obligatory to achieve φ regarding the utility not lower than 0, $O_a^0\varphi$ states that this agent a's obligation of φ satisfies the lowest moral standard or evaluation.

Example 1. Ann (a) needs to send a message to her logistics at the current state w, which contains personal information of her customer. As a responsible seller, Ann considers the utility of "telling the logistics the delivery address" (q) and without "exposing the full name of the customer" (p) to be 0.8, and the utility of telling the delivery address with the full name to be 0.2.

We can model the above scenario by the 0-rank deontic model $M = (W, R, V)$ such that:

- $W = \{w, u, v\}$;
- $R_a(w, u) = 0.8$, $R_a(w, v) = 0.2$, and $R_a(w, w) = R_a(u, u) = R_a(v, v) = R_a(u, w) = R_a(v, w) = R_a(u, v) = R_a(v, u) = 0$;
- $V(p) = \{v\}$ and $V(q) = \{u, v\}$.

The model M is illustrated below:



The following statements are true at state w:

- $M, w \models O_a^{0.1} q \wedge \neg O_a^{0.1} p \wedge \neg O_a^{0.1} \neg p$: "By demanding a minimal utility 0.1, agent a is able to achieve q but neither p nor $\neg p$";
- $M, w \models O_a^{0.5}(\neg p \wedge q)$: "By demanding a minimal utility 0.5, agent a is able to achieve $\neg p$ and q".

By moral standard 0.1 at w, agent a is obligatory to let the logistics know the delivery address, but she is not obligatory not to tell the full name to the logistics. Under the moral standard 0.5, however, agent a is obligatory to protect the customer's privacy, in the sense of not exposing the customer's full name.

We introduce an axiomatization $\mathbf{D}(\rho)$ for the deontic logic shown in Fig. 6. Using a similar method to those in the previous sections we can show that $\mathbf{D}(\rho)$ is sound and complete with respect to the class of all ρ-rank deontic models.

(**PC**)	all instances of propositional tautologies
(UM)	$O_a^{r_1}\varphi \to O_a^{r_2}\varphi$, if $r_1 \leq r_2$
(K)	$O_a^r(\varphi \to \psi) \to (O_a^r\varphi \to O_a^r\psi)$
(D)	$\neg(O_a^\rho\varphi \wedge O_a^\rho\neg\varphi)$
(T^0)	$O_a^0\varphi \to \varphi$
(5^0)	$\neg O_a^0\varphi \to O_a^0\neg O_a^0\varphi$
(A^0)	$O_a^0\varphi \leftrightarrow O_b^0\varphi$
(Tr^1)	$\varphi \to O_a^1\varphi$
(MP)	from φ and $\varphi \to \psi$, infer ψ
(N)	from φ infer $O_a^r\varphi$

Fig. 6. The axiom system $\mathbf{D}(\rho)$.

5 Discussion and Conclusion

We introduced a type of multi-agent weighted modal logics with the partially ordered weights expressing the degrees of epistemic or deontic capability. Such logics provide a general framework for extending standard modal logics with a measurement of the quantity or quality of certain interpretations of necessity, such as knowledge and obligation. By applications in epistemic and deontic contexts, we find the framework flexible for enforcing frame conditions. We also introduced sound and complete axiomatizations for these logics.

The weighted epistemic logic we introduced gets close to [22], due to the practice of measuring similarity numerically using a definition that is essentially convertible to a metric distance. But we hold different opinions towards the use of the weighted models, leading to a different logic. We are interested in adopting more sophisticated conditions of similarity metric, see, e.g., [4].

For future work we are also interested in incorporating group notions of knowledge. Possible extensions in this strand also include the logical dynamics. For example, we can extend the language with the following two sets of formulas:

(1) $\langle a : r\rangle\varphi$ and $\langle a\rangle\varphi$
(2) $\langle\langle a : \psi : s\rangle\rangle\varphi$ and $\langle\langle a\rangle\rangle\varphi$

The formula $\langle a : r\rangle\varphi$ says that "a adjusts its capability to r, and φ is the case", and $\langle a\rangle\varphi$ that "there is an r so that $\langle a : r\rangle\varphi$". The formula $\langle\langle a : \psi : s\rangle\rangle\varphi$ is intended to express that "all ψ-worlds become alike to the degree s for a, and φ is the case", and $\langle\langle a\rangle\rangle\varphi$ that "there are ψ and s so that $\langle\langle a : \psi : s\rangle\rangle\varphi$". The dynamic operators from these types of formulas clearly have a link to the logical characterizations of *knowability* [1,21]. Yet the framework is sufficiently flexible for reasoning about knowability and *forgetability/degeneration* (in the case when r decreases or s increases) w.r.t. *an agent*, via *proactive* (first set) or *passive* (second set) manners. A bare quantification of actions, by a $\Diamond$-operator, can also be introduced to construct formulas such as $\Diamond K_a\varphi$ ("there is a way to make a know φ). Moreover, there is also a third set that is of interest:

(3) $\langle a : b\rangle\varphi$: "$a$ reaches b's capability and φ is the case."

This type of operators may be understood as learning or influencing.

It is natural to compare our deontic logic of weighted capability with other existing deontic logics. Deontic STIT-logic [17] is a modal logic of action, which highlights the aspects of action consequences and their temporal dimensions in the models. By contrast, our deontic models focuses on agent's weighing ability. It is interesting in comparing them in the application of decision-making [2,12,16]. Further, our weighted modality is different from the preferential modality of conditional obligation. Deontic logic of preference (like dyadic deontic logic [10,23] and deontic preference logic [11,28]) captures obligations in terms of maximality or optimality in the Lewis' tradition [19]. This is not the case here. We leave these comparisons in future work.

We have considered the weights in the deontic context as positive values (e.g. utilities). This can also be extended to other possible evaluations. For instance,

in the discussion of norms violation [3,8,27], the concept of deontic capability can be seen as degrees of loss recognition after certain norms are impaired. It can be used to evaluate a moral agent's ability of having compensation to cover the loss.

Our framework has its potential to be applied in more disciplines, and we leave this for future work. One may also consider adding weights to each of the states, modeling some kind of importance or quality of each of them. So far we find little work done in this direction, and it can be an opportunity for a comparison with existing research strands.

Acknowledgments. We would like to thank the two referees of LORI 2021 for their valuable remarks and comments. Huimin Dong acknowledges funding support by the National Social Science Fund of China (20CZX051) and Yì N. Wáng acknowledges funding support by the National Social Science Fund of China (18ZDA290, 20&ZD047).

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